

# Savings-heuristic algorithm for Travelling-salesman problem

- 1. First, all the nodes are sorted in ascending order of total edge-costs from each node. (sum of all column-values in a row of the cost-matrix.)
- 2. Then, the savings-heuristic algorithm is run iteratively, using different nodes as the starting-node.
- 3. The lowest tour-cost valid-tour generated from all the iterations, is found.

The initial iterative executions of the algorithm use the sorted nodes-list for starting-node. But then in subsequent iterations, it sometimes uses randomly-selected starting-nodes to introduce an element of exploration in the search-algorithm strategy.

In certain iterations, the starting-node is also selected from the middle or bottom of the sorted nodes-list.

The same savings-heuristic algorithm works for both euclidean and non-euclidean travelling-salesman problems.

A **maximum-iterations-limit (150)** and **maximum-execution-time-limit (50 seconds)** are also set in the script, for handling huge input data-sets and test-cases. So that the program stops execution with atleast one valid-tour solution, and doesn't keep running for too long.

## Test-run results for the 8 test-cases given in the validator for travelling-salesman problem.

| Test-case runs                         | 0, euc_10              | 1, non-euc_10        | 2, euc_25            | 3, non-euc_25        | 4, euc_50             | 5, non-euc_50       | 6, euc_100           | 7, non-euc_100        |
|----------------------------------------|------------------------|----------------------|----------------------|----------------------|-----------------------|---------------------|----------------------|-----------------------|
| 1. 24-8-2026                           | 655.6751331590185      | 768.988361117        | 964.812249689        | 1609.268829199       | 1340.460459118        | 3109.62367178       | 1814.262870419       | 5688.3586401872       |
| 2. 26-8-2026 11:47                     | 655.6751331590184, (7) | 754.71076386199, (1) | 919.082356725, (17)  | 1726.442756836, (17) | 1319.92637077, (17)   | 3147.410918719, (6) | 1791.15178762, (85)  | 5746.500427348, (69)  |
| 3. 26-8-2026 11:52                     | 655.6751331590184, (5) | 693.48129862649, (2) | 845.0878576714, (11) | 1576.720440208, (9)  | 1223.4319945288, (48) | 2934.84797799, (8)  | 1699.094389369, (88) | 5688.3586401872, (20) |
| 4. 26-8-2026 11:53                     | 655.6751331590184, (5) | 673.7787665244, (9)  | 845.0878576714, (11) | 1584.7737918, (4)    | 1241.6814249, (20)    | 2995.65760728, (33) | 1736.687648228, (34) | 5635.42056207, (0)    |
| 5. 26-8-2026 11:54                     | 655.6751331590184, (5) | 673.7787665244, (9)  | 787.5752663189, (24) | 1576.720440208, (9)  | 1223.4572162, (29)    | 2934.84797799, (8)  | 1707.964491164, (59) | 5591.8855703, (3)     |
| 6. 26-8-2026 12:16; max-iterations=50  | 655.6751331590184, (5) | 673.7787665244, (9)  | 787.5752663189, (24) | 1576.720440208, (9)  | 1196.460993216, (26)  | 2934.84797799, (8)  | 1693.69099236, (22)  | 5622.59455523, (81)   |
| 7. 26-8-2026 12:17; max-iterations=100 | 655.6751331590184, (5) | 673.7787665244, (9)  | 787.5752663189, (24) | 1576.720440208, (9)  | 1196.460993216, (26)  | 2934.84797799, (8)  | 1686.784307951, (99) | 5591.8855703, (3)     |

The starting-node selected for each tour-generation in the above results, is mentioned inside paranthesis () next to each tour-cost.

It is observed in the test-runs listed above, that the lowest-cost valid-tours are found after greater number of iterations on different starting-nodes.

The best and lowest-cost valid-tours for all test-cases are found in the final test-run (**No. 7**) listed above, where the savings-heuristic algorithm was iteratively run with all given nodes as the starting-node in each test-case. (The maximum-iteration limit was set to 100 in the final test-run listed above.)

These results indicate that the node with the lowest total-edge-cost to all other nodes in a t.s.problem, ( node closest to the geometric-center of all the node-locations in euclidean t.s.p. ), is not necessarily the best starting-node for generating the lowest-cost valid-tour using the savings-heuristic algorithm.

### Savings-heuristic algorithm logic :

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1. After the starting-node is selected, all the edges between all the other nodes are sorted in decreasing-order of **edge-cost savings** compared to edges with the starting-node.

**(cost-savings of edge i-to-j) = (cost of i-to-startnode) + (cost of startnode-to-j) - (cost of i-to-j)**

2. Then the valid-tour is generated by adding the highest cost-saving edges one by one, without creating sub-loops or closing the tour prematurely.

3. The total cost of the generated valid-tour is compared to find the minimum-cost valid-tour for each test-case.